Sorting System

How to build automatic storage with item frames on chests.

The chest types

Place an **item frame on a chest** and put one of these items in the frame. The item decides what the chest does.



Input chest

Frame holds an **Eye of Ender**

Anything you put in here is sorted **instantly**. It works with items dropped in by hand, hoppers and droppers.



Receiver chest

Frame holds a **Name Tag** named `#code`

Collects every item with *code* in its name. For example `#Sapling` collects all saplings.



Group chest

Frame holds a **Name Tag** named `.group`

Collects every item of a **group** chosen by the operators, for example `.Redstone` for repeaters, comparators, pistons and more.



Overflow chest

Frame holds a **Carrot on a Stick**

Gets the leftovers: items with no matching name tag, and items whose tagged chests are full.

Setting it up

1. **Place the input chest.** Put an item frame on one of its sides (sneak and right-click the chest) and put an **Eye of Ender** in the frame.
2. **Name a Name Tag in an anvil** so it starts with `#`, for example `#Sapling`, `#Log` or `#Diamond`.
3. **Place the receiver chests** within **128 blocks** of the input chest. Put a frame on each one with its named tag inside.
4. **Place at least one overflow chest** with a **Carrot on a Stick** in its frame, so unmatched items have somewhere to go.
5. **Drop items into the input chest.** They are sent out right away.

> **Without an overflow chest**, items that don't match any tag stay in the input chest. Nothing is lost or deleted.

Naming the tags

The text after `#` is looked up inside the item's name. Upper and lower case don't matter.

Tag	Collects
`#Sapling`	Oak Sapling, Birch Sapling, Cherry Sapling, ... (every sapling)
`#Oak Sapling`	Only Oak Sapling (Dark Oak Sapling too, since its name contains "Oak Sapling")
`#Log`	Every log, including stripped logs
`#Diamond`	Diamond, Diamond Block, Diamond Ore, Diamond Sword, Diamond Pickaxe, ...
`#Seeds`	Wheat Seeds, Pumpkin Seeds, Melon Seeds, Beetroot Seeds, ...

> **Use the English item names.** If your game is set to another language, use the English name or the item ID instead. Press **F3 + H** to show item IDs when you hover over items. For example, `#oak_sapling` works the same as `#Oak Sapling`. Items renamed in an anvil are sorted by their **original** name.

Where each item goes

1. Best # match The longest matching `#` tag, then the closest chest. → **2. Other # matches** If that chest is full, the next matching chest. → **3. Its group** Chests with the item's `.group`, nearest first. → **4. Stock** Chests with a `+` tag, same order. → **5. Overflow** Chests with a Carrot on a Stick, nearest first. → **6. Stays put** If everything is full, the item stays in the input chest.

Example: you have a `#Oak` chest and a `#Sapling` chest. An Oak Sapling matches both, and it goes to `#Sapling` because that tag is longer. Oak Logs and Oak Planks go to `#Oak`.

Names beat groups: with a `#Repeater` chest and a `.Redstone` chest, repeaters go to `#Repeater`; the rest of the Redstone group goes to `.Redstone`.

Shelves and stock

A chest with `.Oak` is a **shelf**. A chest with the same tag and a plus in front, `+.Oak`, is its **stock**: it only gets items when the shelves are full, and it **refills the shelf** as soon as you take something and close it (or a hopper pulls from it). The stock can be up to 64 blocks away, for example on the floor above. When you close a chest of the system it is tidied: stacks merged and items in the creative inventory order, like Inventory Profiles Next.

Item groups

A group is a list of items picked by hand, for things whose names have nothing in common (redstone parts, farm drops, ores, ...). Each item is in at most one group. The operators edit the groups in a web page: they type `/gems web` in game and open the link. To use a group, rename

a Name Tag to a dot followed by the group name, for example `.Redstone` (upper and lower case don't matter), and put it in a frame on the chest. Changes to a group apply right away.

Good to know

- **Range:** receiver and overflow chests must be within **128 blocks** of the input chest.
- **Stay nearby:** far-away chests in areas that aren't loaded are skipped. Their items go to the overflow chest instead.
- **Double chests** work. The frame can go on either half.
- **Any chest works**: trapped chests, copper chests (oxidized or waxed too) and barrels, for input, tags and overflow. Glow item frames work too. The frame can go on any side of the chest, including the top.
- You can have **several input chests**, and several chests with the same tag. When one fills up, the next one takes over.
- Changes to frames take up to **3 seconds** to take effect.
- Don't put an Eye of Ender on a receiver chest: that turns it into an input chest, and nothing gets sorted into it.
- **Nothing sorts?** Check that the name tag starts with `#` (or `.` for a group) and was renamed in an anvil, that the frame is on the chest itself, and that the chest is within 128 blocks. For a group, ask an operator whether the group exists.